

Program : Diploma in Robotic Process Automation	
Course Code : 3303	Course Title: Introduction to Robotic Process Automation
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To understand the basic concepts of automation.
- To learn about different types of bots.
- To learn and understand Robotic Process Automation and their working.
- To learn about different RPA tools.
- To understand the challenges & risks when implementing RPA.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic programming skills in C language.		Introduction to Computation and Programming	2

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concepts of RPA and its architecture.	18	Understand
CO2	Understand the robotic control flow architecture and familiarize the tool used in RPA.	21	Understand

CO3	Understand the different stages in the RPA life cycle.	8	Understand
CO4	Understand the working of RPA with existing systems and the features of various tools used in RPA.	11	Understand
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3				3	3	
CO2	3				3	3	
CO3	3				3	3	
CO4	3			3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concepts of RPA and its architecture.		
M1.01	Define automation, scope and techniques of automation	1	Understanding
M1.02	Describe processes and flowcharts	2	Understanding
M1.03	Illustrate the concepts and benefits of RPA	1	Understanding
M1.04	Differentiate traditional automation and Robotic Process Automation	1	Understanding
M1.05	List and explain RPA services	1	Understanding
M1.06	Describe the advantages and disadvantages of RPA	1	Understanding
M1.07	Explain bots and types of bots	2	Understanding
M1.08	Describe the architecture of RPA	1	Understanding.
M1.09	Explain applications under RPA	1	Understanding

M1.10	Describe RPA tools.	1	Understanding.
M1.11	Explain RPA platform	1	Understanding.
M1.12	Explain RPA execution infrastructure.	1	Understanding.
M1.13	Explain programming elements in RPA	1	Understanding
M1.14	Describe recorder.	1	Understanding
M1.15	Explain development studio and plug-in/extension	1	Understanding
M1.16	Describe bot runner and control center.	1	Understanding

Contents:

Fundamentals of RPA: - What is automation? Scope of automation, techniques of automation, processes and flow charts. RPA - history of RPA, benefits of RPA, Automation vs RPA, RPA services, advantages and disadvantages. Bots-types of bots.

RPA Architecture – Applications under Robotic Process Execution, RPA tools, RPA platform, RPA execution infrastructure.

Programming elements in RPA – Recorder, Development studio, Plug-in/Extension, Bot runner, control center.

CO2:	Understand the robotic control flow architecture and familiarize the tool used in RPA.		
M2.01	Explain RPA Development methodologies	1	Understanding
M2.02	Illustrate the Robotic control flow architecture with examples	2	Understanding
M2.03	Describe RPA business case	1	Understanding.
M2.04	Explain Process Design Document/Solution Design Document	1	Understanding
M2.05	Explain the risks and challenges with RPA	1	Understanding.
M2.06	Explain the different types of variables and attributes with examples	3	Understanding.
M2.07	Describe the control flow statements with examples	4	Understanding.
M2.08	Illustrate the control flow activities with examples	4	Understanding
M2.09	Illustrate basic and desktop recording	1	Understanding
M2.10	Explain web recording	1	Understanding

M2.11	Describe input/output methods	1	Understanding
M2.12	Describe screen scraping with example	1	Understanding
	Series Test – I	1	
Contents: Standardization of processes - RPA Development methodologies -Robotic control flow architecture - RPA business case - Process Design Document/Solution Design Document - Industries best suited for RPA - Risks & Challenges with RPA. Introduction to UiPath –Variables (Managing Variables, Naming Best Practices, The Variables Panel, Generic Value Variables, Text Variables, True or False Variables, Number Variables, Array Variables, Date and Time Variables, Data Table Variables), Control Flow (If Else Statements, Loops, Advanced Control Flow, Sequences, Flowchart), Control Flow Activities (The Assign Activity, The Delay Activity, The Do While Activity, The If Activity, The Switch Activity, The While Activity, The For Each Activity, The Break Activity), Arguments, Recording (Recording Introduction, Basic and Desktop Recording, Web Recording, Input/Output Methods). Screen scraping.			
CO3:	Understand the different stages in the RPA life cycle.		
M3.01	Explain the life cycle of RPA-discovery phase	2	Understanding
M3.02	Describe solution design phase and development phase	1	Understanding
M3.03	Describe user acceptance test (UAT) and deployment and maintenance phase.	1	Understanding.
M3.04	Explain execute bots phase.	2	Understanding.
M3.05	Describe RPA implementation.	2	Understanding
Contents: RPA life cycle – Discovery phase, solution design phase, development phase, user acceptance test (UAT), deployment and maintenance phase, execute bots. RPA implementation.			
CO4:	Understand the working of RPA with existing systems and the features of various tools used in RPA.		
M4.01	Explain how RPA works?	1	Understanding
M4.02	Describe how RPA works with existing systems? back end and front end integrations.	1	Understanding
M4.03	Explain the working of AI powered RPA.	2	Understanding
M4.04	List RPA tools-benefits of UiPath tool	2	Understanding
M4.05	Explain features of Blue Prism.	1	Understanding

M4.06	Explain features of Automation Anywhere tool.	1	Understanding
M4.07	Explain features of Pega and WorkFusion	1	Understanding
M4.08	Describe the core functionalities of RPA tools	1	Understanding
M4.09	Describe the parameters for choosing the right RPA tool.	1	Understanding
	Series Test – II	1	

Contents:

Working of RPA-with the existing systems,-back end and front end integrations, working of AI powered RPA. RPA tools-UiPath, Blue Prism, Automation Anywhere, Pega, WorkFusion,- Core Functionalities of RPA tools. Parameters for choosing the right RPA tool.

Text / Reference

T/R	Book Title/Author
T1	Alok Mani Tripathi, “ Learning Robotic Process Automation ”, Packt Publishing, 2018
R1	Frank Casale , Rebecca Dilla, Heidi Jaynes , Lauren Livingston, “ Introduction to Robotic Process Automation: a Primer ”, Institute of Robotic Process Automation,1st Edition 2015.
R2	Richard Murdoch, Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant ”, Independently Published, 1st Edition 2018.
R3	Srikanth Merianda,” Robotic Process Automation Tools, Process Automation and their benefits: Understanding RPA and Intelligent Automation ”, Consulting Opportunity Holdings LLC, 1st Edition 2018.
R4	Lim Mei Ying, “ Robotic Process Automation with Blue Prism Quick Start Guide: Create software robots and automate business processes ”, Packt Publishing, 1st Edition 2018.

Online Resources

Sl.No	Website Link
1	https://www.w3schools.com/html/
2	https://www.w3schools.com/xml/
3	https://www.udemy.com/course/robotic-process-automation/
4	https://www.javatpoint.com/rpa
5	https://intellipaat.com/blog/tutorial/rpa-tutorial/
6	https://dev.tutorialspoint.com/uiopath/uiopath_tutorial.pdf